# 1.3 The Triangle Inequality

The modulus of Section 1.2 measures a complex number by a real one. Picture $z = a + bi$ as the point $(a, b)$ of the plane: by Pythagoras, $|z| = \sqrt{a^2 + b^2}$ is its distance from the origin, and $|z - w|$ is the distance between the points $z$ and $w$. In a triangle, one side is never longer than the other two together. This section proves the algebraic form of that fact, $|z + w| \leq |z| + |w|$, from the definitions alone. The picture motivates it; no step of the proof uses it.

## 1.3.1 Why the old proof does not transfer

For an ordered field, *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$, Chapter 4 proved $|x + y| \leq |x| + |y|$ by trapping (Number Systems: From \(\nN\) to \(\nR\), Proposition 4.1.3 (e)). Each of $x$ and $y$ lies between $\pm$ its absolute value, $-|x| \leq x \leq |x|$ (part (c) there); adding the two traps traps $x + y$ between $\pm(|x| + |y|)$, and part (d) reads that as the inequality.

That route needs $x \leq |x|$: a comparison between $x$ itself and a number. For a complex $z$, "$z \leq |z|$" compares a complex number with a real one, and $\mathbb{C}$ has been given no order in which to make the comparison. Section 1.4 shows that no order compatible with the arithmetic of $\mathbb{C}$ exists (Proposition 1.4.3). So the proof must compare **real** numbers only.

Two kinds of real number are at hand. Moduli are real, and so are real parts. The device that connects them to the arithmetic of $\mathbb{C}$ is the identity $|u|^2 = u\overline{u}$ of Theorem 1.2.2 (c). A modulus is a square root, and square roots do not add well; its square is a product, and products expand. So we will compare squares and then come back.

**Comparing squares.** Coming back from squares to the numbers needs one rule about the real numbers. Let $p, q \in \mathbb{R}$ with $p \geq 0$ and $q \geq 0$. Then

$$p \leq q \iff p^2 \leq q^2, \qquad p = q \iff p^2 = q^2,$$

and likewise with $<$ on both sides. The strict form is Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (g). For the first equivalence: if $p < q$, then $p^2 < q^2$ by (g), and if $p = q$ the squares agree. Conversely, if $p^2 \leq q^2$ and yet $q < p$, then (g) gives $q^2 < p^2$, which contradicts $p^2 \leq q^2$ by trichotomy (O1); so $p \leq q$. The second equivalence follows from the first, applied in both directions, and trichotomy. In words: *for non-negative reals, squaring neither creates nor destroys a comparison.* Both hypotheses $p \geq 0$ and $q \geq 0$ matter: $(-2)^2 > 1^2$, and yet $-2 < 1$.

## 1.3.2 The real part is at most the modulus

The single estimate the proof needs is that a real part never exceeds the modulus. It comes with two companions, which say that the modulus and the "taxicab size" $|\mathrm{Re}\, u| + |\mathrm{Im}\, u|$ are comparable.

**Lemma 1.3.1. Real and Imaginary Parts against the Modulus**

For every $u \in \mathbb{C}$,

$$|\mathrm{Re}\, u| \leq |u|, \qquad |\mathrm{Im}\, u| \leq |u|, \qquad |u| \leq |\mathrm{Re}\, u| + |\mathrm{Im}\, u|.$$

In particular $\mathrm{Re}\, u \leq |u|$.

Here $|\operatorname{Re} u|$ is the absolute value of a real number. By Proposition 1.2.1 it is the same whether it is read as the absolute value of the ordered field $\mathbb{R}$ from *Number Systems: From $\mathbb{N}$ to $\mathbb{R}$*, Chapter 4 (Number Systems: From \(\nN\) to \(\nR\)), Definition 4.1.1) or as the modulus of a complex number that happens to be real, so the lemma has only one reading.

*Idea.* Every quantity in the lemma is a non-negative real, so compare squares. The square of $|u|$ is $x^2+y^2$, and the squares of $|x|$ and $|y|$ are $x^2$ and $y^2$, each at most the sum; the square of $|x|+|y|$ is the sum plus the non-negative cross term $2|x||y|$.

*Proof.* Write $u = x + yi$ with $x, y \in \mathbb{R}$. By Definition 1.2.1 and the defining property of the square root (Number Systems: From \(\nN\) to \(\nR\), Proposition 6.1.4), $|u| \geq 0$ and $|u|^2 = x^2 + y^2$.

Since $y^2 \geq 0$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (e) if $y \neq 0$; $0^2 = 0$ otherwise), adding $x^2$ gives $x^2 \leq x^2 + y^2$ by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (b). Also $|x|^2 = x^2$ and $|x| \geq 0$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 4.1.3 (a), (b)). Hence $|x|^2 \leq |u|^2$, with both $|x|$ and $|u|$ non-negative, and comparing squares gives $|x| \leq |u|$. The same argument with the roles of $x$ and $y$ exchanged gives $|y| \leq |u|$. Then $\operatorname{Re} u = x \leq |x|$ by Number Systems: From \(\nN\) to \(\nR\), Proposition 4.1.3 (c), so $\operatorname{Re} u \leq |u|$.

For the last inequality, expand with Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (d):

$$\begin{aligned}\big(|x| + |y|\big)^2 &= |x|^2 + 2|x||y| + |y|^2 \\ &= x^2 + y^2 + 2|x||y| \geq x^2 + y^2 = |u|^2,\end{aligned}$$

where the inequality holds because $|x||y| \geq 0$ as a product of non-negative reals, so $2|x||y| \geq 0$, and Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (b) adds it. Both $|x|+|y|$ and $|u|$ are non-negative, so comparing squares gives $|u| \leq |x| + |y|$. This proves the lemma. ∎

For $u = 3 - 4i$ the three inequalities read $3 \leq 5$, $4 \leq 5$ and $5 \leq 7$. For $u = 2$ they read $2 \leq 2$, $0 \leq 2$ and $2 \leq 2$: a real $u$ makes the first and third equalities. The inequality $\operatorname{Re} u \leq |u|$ is strict for $u = -1$, where it reads $-1 \leq 1$; the absolute value inside the first inequality is what makes it sharp for real $u$ of either sign.

*Quick check.* Find a non-zero $u \in \mathbb{C}$ with $\operatorname{Re} u = 0$. What do the three inequalities of Lemma 1.3.1 say for it?

*Solution.* For example $u = 2i$, with $\operatorname{Re} u = 0$, $\operatorname{Im} u = 2$ and $|u| = \sqrt{0+4} = 2$. The inequalities read $0 \leq 2$, $2 \leq 2$ and $2 \leq 0 + 2$. Any $u = yi$ with $y \neq 0$ works the same way: $|u| = |y|$, so the second and third inequalities are equalities.

## 1.3.3 The triangle inequality

We can now state the theorem. Its proof is the first full run of the move this chapter is built around.

**Theorem 1.3.2. Triangle Inequality for Complex Numbers**

For all $z, w \in \mathbb{C}$,

$$|z + w| \leq |z| + |w|.$$

*Try it first (ten minutes): show $|z + w|^2 \leq (|z| + |w|)^2$ by expanding both sides with $|u|^2 = u\overline{u}$, and note which term you cannot yet bound.*

*Idea.* Expanding the left side with $|u|^2 = u\overline{u}$ and conjugation of sums gives $|z|^2 + z\overline{w} + \overline{z}w + |w|^2$; the right side is $|z|^2 + 2|z||w| + |w|^2$. Everything matches except the cross term $z\overline{w} + \overline{z}w$, and an attempt usually stops there: it looks like a complex number, and complex numbers cannot be compared with $2|z||w|$. The way out is to notice that $\overline{z}w$ is the conjugate of $z\overline{w}$, so the cross term is $u + \overline{u} = 2\operatorname{Re} u$ with $u = z\overline{w}$: a **real** number after all. A real part is at most the modulus, and $|z\overline{w}| = |z||w|$. Squares compare, so the moduli do.

*Before reading on: where is $\operatorname{Re} u \leq |u|$ used, and why may we compare squares?*

*Proof.* Let $z, w \in \mathbb{C}$. By Theorem 1.2.2 (c), then (a), then Proposition 1.1.2 (c) together with (a) and (b) (distributivity, commutativity, associativity),

$$\begin{aligned} |z + w|^2 &= (z + w)\,\overline{(z + w)} = (z + w)(\overline{z} + \overline{w}) \\ &= z\overline{z} + z\overline{w} + \overline{z}w + w\overline{w}. \end{aligned}$$

By Theorem 1.2.2 (b) and (a), $\overline{z\overline{w}} = \overline{z}\,\overline{\overline{w}} = \overline{z}w$. So with $u \coloneqq z\overline{w}$, the two middle terms are $u + \overline{u} = 2\operatorname{Re} u$, by Theorem 1.2.2 (c). Using (c) once more for $z\overline{z}$ and $w\overline{w}$,

$$|z + w|^2 = |z|^2 + 2\operatorname{Re}(z\overline{w}) + |w|^2.$$

Every term here is real, and sums and products of real numbers are the same in $\mathbb{C}$ as in $\mathbb{R}$ (Proposition 1.1.1), so the inequalities below take place in the ordered field $\mathbb{R}$. By Lemma 1.3.1, $\operatorname{Re}(z\overline{w}) \leq |z\overline{w}|$, and by Theorem 1.2.2 (d), $|z\overline{w}| = |z|\,|\overline{w}| = |z|\,|w|$. Adding this inequality to itself ($2a = (1 + 1)a = a + a$ by distributivity) and then adding $|z|^2 + |w|^2$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (b) twice), and expanding the square with Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (d),

$$|z + w|^2 \leq |z|^2 + 2|z||w| + |w|^2 = \big(|z| + |w|\big)^2.$$

Both $|z+w|$ and $|z|+|w|$ are non-negative real numbers, so comparing squares gives $|z+w| \leq |z| + |w|$. This proves the triangle inequality.

■

The proof answers its prompt in two lines. $\operatorname{Re} u \leq |u|$ is used once, to bound the cross term $2\operatorname{Re}(z\overline{w})$ by $2|z\overline{w}|$. Comparing squares is legitimate because the two quantities being compared, $|z + w|$ and $|z| + |w|$, are both non-negative; without that, a smaller square would say nothing

about the numbers.

This route has a name in this chapter: **square the modulus**. *The modulus is a square root, and square roots are hard to handle; its square $|u|^2 = u\overline{u}$ is a product, so square, expand with conjugates, bound or compare the real numbers you get, and take the non-negative square root back.* Section 1.2 used the same move once already, for $|zw| = |z||w|$ (Theorem 1.2.2 (d)). The linear algebra volumes take the same route, square, expand, bound the cross term, for the length of a vector in an inner product space.

*Warning.* The square of the modulus is $u\overline{u}$, not $u^2$. Expanding $|z+w|^2$ as $(z+w)^2 = z^2+2zw+w^2$ gives nonsense: for $z = w = i$, $(z+w)^2 = (2i)^2 = -4$, while $|z+w|^2 = |2i|^2 = 4$. The cross term of the correct expansion is $z\overline{w} + \overline{z}w$, and only that form is real.

*Remark.* The inequality can be strict: for $z = 1$ and $w = -1$, the left side is $|0| = 0$ and the right side is 2. For real $z, w$, every modulus in the statement is the absolute value of *Number Systems: From $\mathbb{N}$ to $\mathbb{R}$*, Chapter 4 (Proposition 1.2.1), and the theorem is Number Systems: From \(\nN\) to \(\nR\), Proposition 4.1.3 (e), reached by a different route.

*Quick check.* Compute $|3+4i| + |-3+4i|$ and $|(3+4i)+(-3+4i)|$, and check them against the triangle inequality.

*Solution.* $|3+4i| = \sqrt{9+16} = 5$ and $|-3+4i| = \sqrt{9+16} = 5$, so the right side is 10. The sum is $8i$, and $|8i| = \sqrt{0+64} = 8$. Indeed $8 \le 10$. In the picture, the two points and $8i$ are three corners of a parallelogram with the origin as the fourth, and the diagonal 8 is shorter than the two sides $5+5$.

## 1.3.4 Three variants

Three consequences are used often enough to be stated once: a lower bound for the modulus of a difference, the inequality for any finite number of terms, and a lower bound for the modulus of a sum.

**Corollary 1.3.3. Variants of the Triangle Inequality**

Let $z, w \in \mathbb{C}$.

(a) *(Reverse triangle inequality.)* $\big||z| - |w|\big| \le |z-w|$.

(b) For every $n \in \mathbb{N}$ with $n \ge 1$ and all $z_1, \dots, z_n \in \mathbb{C}$, $\left|\sum_{k=1}^{n} z_k\right| \le \sum_{k=1}^{n} |z_k|$.

(c) $|z| - |w| \le |z+w|$.

In (a), the outer bars are the absolute value of the real number $|z| - |w|$. In words: (a) says that two points at distance $|z-w|$ from each other have distances from the origin that differ by at most that much; (b) says that a path of $n$ steps is at least as long as the straight line from

start to end; (c) says that adding $w$ can lower the modulus by at most $|w|$.

*Idea.* For (a), split $z$ as $(z-w)+w$ and apply the theorem; that bounds $|z|-|w|$ from above, and swapping $z$ and $w$ bounds it from below. For (b), add one term at a time. For (c), apply (a) to $z$ and $-w$.

*Proof.*

(a) Since $z=(z-w)+w$, Theorem 1.3.2 gives $|z|\le|z-w|+|w|$, so $|z|-|w|\le|z-w|$. Exchanging $z$ and $w$ gives $|w|-|z|\le|w-z|$. Moreover $w-z=(-1)(z-w)$, so by Theorem 1.2.2 (d), $|w-z|=|-1|\,|z-w|=|z-w|$, where $|-1|=1$ by Proposition 1.2.1. Let $t:=|z|-|w|\in\mathbb{R}$. We have shown $t\le|z-w|$ and $-t\le|z-w|$, that is, $-|z-w|\le t\le|z-w|$: if $-t<|z-w|$, then $-|z-w|<-(-t)=t$ by Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (a) and Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (a), and if $-t=|z-w|$, then $-|z-w|=-(-t)=t$. By Number Systems: From \(\nN\) to \(\nR\), Proposition 4.1.3 (d), $|t|\le|z-w|$.

(b) By induction on $n\ge1$ (Logic and Proof, Axiom 4.2.1, through Number Systems: From \(\nN\) to \(\nR\), Corollary 1.7.3 (b), applied to the statement for $n=m+1$, $m\in\mathbb{N}$), with the sums defined by appending the last term, as for real sums (Sets, Relations, and Cardinality, Definition 2.3.1). For $n=1$, both sides are $|z_1|$. Suppose the inequality holds for $n$, and let $z_1,\dots,z_{n+1}\in\mathbb{C}$. Then

$$\begin{aligned}\left|\sum_{k=1}^{n+1}z_k\right|&=\left|\sum_{k=1}^{n}z_k+z_{n+1}\right|\le\left|\sum_{k=1}^{n}z_k\right|+|z_{n+1}|\\&\le\sum_{k=1}^{n}|z_k|+|z_{n+1}|=\sum_{k=1}^{n+1}|z_k|,\end{aligned}$$

by Theorem 1.3.2, then the induction hypothesis with Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (b).

(c) Apply (a) to $z$ and $-w$. Since $z-(-w)=z+w$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (a)) and $|-w|=|-1||w|=|w|$ as in (a), it reads $\big||z|-|w|\big|\le|z+w|$. By Number Systems: From \(\nN\) to \(\nR\), Proposition 4.1.3 (c), $|z|-|w|\le\big||z|-|w|\big|$. This proves the corollary. ∎

Part (b) with $n=2$ is the theorem itself, so the corollary contains it. *Algebraic Preliminaries*, Chapter 4 uses (b) and (c) to show that a polynomial expression of degree at least 1 is large in modulus when $|z|$ is large, a step towards the Fundamental Theorem of Algebra.

*Quick check.* Take $z=5$ and $w=3i$. Compute both sides of (a) and of (c).

*Solution.* $|z|=5$ and $|w|=3$, so $|z|-|w|=2$. For (a): $|z-w|=|5-3i|=\sqrt{34}$, and $2\le\sqrt{34}$ because $2^2=4\le34$ (comparing squares). For (c): $|z+w|=|5+3i|=\sqrt{34}$ too,

and again $2 \le \sqrt{34}$. Both bounds are far from sharp here; (a) is an equality when $w = 0$, for instance.

## 1.3.5 Exercises

### 1.3.5.1 A. Check your understanding

**Exercise. A1**

These questions are about the proof of Theorem 1.3.2.

(a) Where in the proof is $\operatorname{Re} u \le |u|$ used, and for which $u$?

(b) Why may the proof pass from $|z+w|^2 \le (|z|+|w|)^2$ back to $|z+w| \le |z|+|w|$?

(c) Run the proof on $z = 3$ and $w = 4i$: compute the cross term $2\operatorname{Re}(z\overline{w})$, then both sides of the squared inequality.

(d) State the idea of the proof in two sentences.

(e) True or false: $|z+w| = |z|+|w|$ for all $z, w \in \mathbb{C}$. Justify your answer.

*Solution.*

(a) Once, to bound the cross term: with $u = z\overline{w}$, $2\operatorname{Re}(z\overline{w}) \le 2|z\overline{w}|$.

(b) Both $|z+w|$ and $|z|+|w|$ are non-negative reals, and for non-negative reals $p \le q$ if and only if $p^2 \le q^2$ (comparing squares, from Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (g) and trichotomy).

(c) $\overline{w} = -4i$, so $z\overline{w} = 3 \cdot (-4i) = -12i$, with real part 0: the cross term is 0. Then $|z+w|^2 = 9 + 0 + 16 = 25$ and $(|z|+|w|)^2 = (3+4)^2 = 49$, and $25 \le 49$. So $|3+4i| = 5 \le 7$.

(d) Square both sides and expand with $|u|^2 = u\overline{u}$: everything matches except the cross term, which is $2\operatorname{Re}(z\overline{w})$. A real part is at most the modulus, and $|z\overline{w}| = |z||w|$, so the squares compare, and so do the non-negative moduli.

(e) False. For $z = 1$ and $w = -1$, $|z+w| = 0$, but $|z|+|w| = 2$.

### 1.3.5.2 B. Practice

**Exercise. B1: The equality case**

Let $z, w \in \mathbb{C}$. Prove that $|z+w| = |z|+|w|$ if and only if $z\overline{w}$ is a non-negative real number, that is, $\operatorname{Im}(z\overline{w}) = 0$ and $\operatorname{Re}(z\overline{w}) \ge 0$.

*Hint: trace the proof of the triangle inequality and ask where an inequality could be strict.*

*Solution.* Let $u := z\overline{w} = x + yi$ with $x, y \in \mathbb{R}$. By the proof of Theorem 1.3.2,

$$|z+w|^2 = |z|^2 + 2x + |w|^2,$$
$$(|z|+|w|)^2 = |z|^2 + 2|u| + |w|^2,$$

the second because $|z||w| = |u|$ (Theorem 1.2.2 (d)). Both moduli are non-negative, so

(comparing squares) $|z+w| = |z| + |w|$ holds if and only if the squares are equal, that is, if and only if $2x = 2|u|$, that is, $x = |u|$.

It remains to show that $x = |u|$ holds exactly when $y = 0$ and $x \geq 0$. If $x = |u|$, then $x \geq 0$, and squaring gives $x^2 = x^2 + y^2$, so $y^2 = 0$ and $y = 0$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (e): a non-zero square is positive). Conversely, if $y = 0$ and $x \geq 0$, then $|u| = \sqrt{x^2} = x$, since $x \geq 0$ and $x^2 = x^2$ (uniqueness in Number Systems: From \(\nN\) to \(\nR\), Proposition 6.1.4). Hence $|z+w| = |z| + |w|$ if and only if $z\overline{w}$ is a non-negative real number.

For instance $z = 1+i$, $w = 2+2i$: $z\overline{w} = (1+i)(2-2i) = 4 \geq 0$, and indeed $|3+3i| = \sqrt{18} = 3\sqrt{2} = \sqrt{2} + 2\sqrt{2} = |1+i| + |2+2i|$. In the picture: equality holds exactly when one of $z, w$ is 0 or they point in the same direction from the origin. The linear algebra volumes prove the version for $n$ terms.

**Exercise. B2: A distance that does not change**

Let $z \in \mathbb{C}$ with $|z| = 1$. Prove that $|z - w| = |1 - \overline{z}w|$ for every $w \in \mathbb{C}$. The first two steps are done:

$$|z-w|^2 = |z|^2 - z\overline{w} - \overline{z}w + |w|^2,$$
$$|1-\overline{z}w|^2 = 1 - z\overline{w} - \overline{z}w + |z|^2|w|^2,$$

both by expanding $u\overline{u}$ with Theorem 1.2.2. *Name the move, then complete it.*

*Solution.* The move is **square the modulus**: compare the squares, which are algebra, and take the non-negative square roots back.

Substitute $|z|^2 = 1$ into both lines. The first becomes $1 - z\overline{w} - \overline{z}w + |w|^2$, and so does the second, since $|z|^2|w|^2 = |w|^2$. So $|z-w|^2 = |1-\overline{z}w|^2$. Both $|z-w|$ and $|1-\overline{z}w|$ are non-negative, and non-negative reals with equal squares are equal (comparing squares). Hence $|z-w| = |1-\overline{z}w|$.

(For the record, the given lines. Conjugation respects differences: $\overline{-v} = \overline{(-1)v} = \overline{-1}\,\overline{v} = -\overline{v}$ by Theorem 1.2.2 (b), Proposition 1.2.1 and Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.2 (c), so with (a), $\overline{u-v} = \overline{u} - \overline{v}$; and $\overline{1} = 1$ by Proposition 1.2.1. Then $(z-w)(\overline{z}-\overline{w}) = z\overline{z} - z\overline{w} - w\overline{z} + w\overline{w}$, and $\overline{1-\overline{z}w} = 1 - z\overline{w}$, so $(1-\overline{z}w)(1-z\overline{w}) = 1 - z\overline{w} - \overline{z}w + (z\overline{z})(w\overline{w})$.)

**Exercise. B3**

Prove that $|z| \leq |\operatorname{Re} z| + |\operatorname{Im} z| \leq \sqrt{2}\,|z|$ for every $z \in \mathbb{C}$, and, for each of the two inequalities, find a non-zero $z$ for which it is an equality.

*Solution.* The first inequality is the last part of Lemma 1.3.1. For the second, write $z = x + yi$. Both sides are non-negative ($\sqrt{2} \geq 0$ by Number Systems: From \(\nN\) to \(\nR\), Proposition 6.1.4), so it suffices to compare squares. The right side squared is $(\sqrt{2})^2|z|^2 = 2(x^2+y^2)$, and the left side squared is $x^2 + 2|x||y| + y^2$, using $|x|^2 = x^2$ (Number Systems:

From \(\nN\) to \(\nR\), Proposition 4.1.3 (b)). Their difference is

$$\begin{aligned} 2(x^2+y^2) - (x^2 + 2|x||y| + y^2) &= |x|^2 - 2|x||y| + |y|^2 \\ &= \big(|x| - |y|\big)^2 \geq 0, \end{aligned}$$

a square in the ordered field $\mathbb{R}$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (e)). So $(|x| + |y|)^2 \leq 2|z|^2$, and $|x| + |y| \leq \sqrt{2}\,|z|$.

Equality: $z = 1$ gives $1 = 1 + 0$ in the first; $z = 1 + i$ gives $1 + 1 = 2 = \sqrt{2} \cdot \sqrt{2}$ in the second.

#### 1.3.5.3 C. Going deeper

**Exercise. C1: Exit check: the triangle inequality**

State and prove the triangle inequality for complex numbers. Your proof must include a proof that $\operatorname{Re} u \leq |u|$ for every $u \in \mathbb{C}$.

*Solution. Statement.* For all $z, w \in \mathbb{C}$, $|z + w| \leq |z| + |w|$.

*The estimate.* Let $u = x + yi$ with $x, y \in \mathbb{R}$. Then $|u|^2 = x^2 + y^2 \geq x^2 = |x|^2$, since $y^2 \geq 0$. Both $|x|$ and $|u|$ are non-negative, so $|x| \leq |u|$ by comparing squares (Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (g) with trichotomy). Hence $\operatorname{Re} u = x \leq |x| \leq |u|$ (Number Systems: From \(\nN\) to \(\nR\), Proposition 4.1.3 (c)).

*The inequality.* Let $z, w \in \mathbb{C}$. By Theorem 1.2.2 (a)–(c),

$$\begin{aligned} |z + w|^2 &= (z + w)(\overline{z} + \overline{w}) = |z|^2 + (z\overline{w} + \overline{z\overline{w}}) + |w|^2 \\ &= |z|^2 + 2\operatorname{Re}(z\overline{w}) + |w|^2. \end{aligned}$$

By the estimate and Theorem 1.2.2 (d), $2\operatorname{Re}(z\overline{w}) \leq 2|z\overline{w}| = 2|z||w|$. Hence $|z+w|^2 \leq (|z| + |w|)^2$. Both $|z+w|$ and $|z|+|w|$ are non-negative, so comparing squares gives $|z+w| \leq |z|+|w|$.

**Marking scheme.** (10 points) - 2: proves $\operatorname{Re} u \leq |u|$, with a reason for passing from squares to moduli. - 3: expands $|z + w|^2$ with conjugates and identifies the cross term as $2\operatorname{Re}(z\overline{w})$, a real number. - 2: bounds the cross term by $2|z||w|$, citing $|z\overline{w}| = |z||w|$. - 1: concludes $|z+w|^2 \leq (|z|+|w|)^2$. - 2: takes square roots, justified by the non-negativity of both sides. A second route, with coordinates $z = a+bi$, $w = c+di$: expanding $(a+c)^2+(b+d)^2$ and isolating the cross term $2(ac + bd)$ earns the 3 points of the second line; proving $ac + bd \leq |z||w|$, for instance from $(ac + bd)^2 \leq (a^2 + b^2)(c^2 + d^2)$, which follows from $(ad - bc)^2 \geq 0$, earns the 2 points of the third line if proved, not if quoted; the last two lines are marked as above. The 2 points for $\operatorname{Re} u \leq |u|$ are still owed on that route, since the problem asks for it. Fatal errors (0 for the step): writing $z \leq |z|$ or any inequality between non-real complex numbers; dropping the conjugate, as in $|u|^2 = u^2$.

**Exercise. C2: Inside the unit disc**

Let $z, w \in \mathbb{C}$ with $|z| < 1$ and $|w| < 1$. Prove that $|z - w| < |1 - \overline{z}w|$. Here is the start of a proof: square both sides and expand. *Name the move, then complete it.*

*Solution.* The move is **square the modulus**. Both sides are non-negative, so it is enough to show $|z-w|^2 < |1-\overline{z}w|^2$ (comparing squares, strict form: Number Systems: From \(\nN\) to \(\nR\), Proposition 2.3.5 (g)). Expanding each as $u\overline{u}$, exactly as in Exercise (B2: A distance that does not change),

$$|z-w|^2 = |z|^2 - z\overline{w} - \overline{z}w + |w|^2,$$
$$|1-\overline{z}w|^2 = 1 - z\overline{w} - \overline{z}w + |z|^2|w|^2.$$

The non-real terms cancel in the difference:

$$\begin{aligned} |1-\overline{z}w|^2 - |z-w|^2 &= 1 - |z|^2 - |w|^2 + |z|^2|w|^2 \\ &= (1-|z|^2)(1-|w|^2). \end{aligned}$$

Since $0 \le |z| < 1$, comparing squares gives $|z|^2 < 1^2 = 1$, so $1-|z|^2 > 0$; likewise $1-|w|^2 > 0$. A product of two positive reals is positive ((O4)). So the difference is positive, $|z-w|^2 < |1-\overline{z}w|^2$, and $|z-w| < |1-\overline{z}w|$.

In particular $1-\overline{z}w \neq 0$. Since $|1-\overline{z}w| = |1-z\overline{w}|$ (a modulus is unchanged by conjugation, Theorem 1.2.2 (d)), the inequality says that $|z-w|/|1-\overline{w}z| < 1$ for all $z, w$ in the open unit disc: the map $z \mapsto (z-w)/(1-\overline{w}z)$ sends the open unit disc into itself, a fact complex analysis builds on.

**Marking scheme.** (10 points) - 2: reduces to the squared inequality, with the non-negativity of both sides as the reason. - 3: expands both squares with conjugates. - 3: factors the difference as $(1-|z|^2)(1-|w|^2)$. - 2: shows both factors positive, from $|z|, |w| < 1$, and concludes strictly. Fatal errors (0 for the step): expanding $|z-w|^2$ as $(z-w)^2$; comparing $z\overline{w}$ with a real number.